CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 5: Playing with Numbers

Topic-wise short notes with solved examples

Numbers in generalised form, number tricks with reversed digits, puzzles in which letters stand for digits, and the tests of divisibility with the reasons behind them.

1 Generalised form

- A two-digit number with tens digit a and units digit b is $10a + b$. It is written ab only as a shorthand, not as a product.
- A three-digit number with digits a, b, c is $100a + 10b + c$.
- Reversing the digits of $10a + b$ gives $10b + a$.

Example 1.1 Write 73 and 508 in generalised form.

SOLUTION

$10 \times 7 + 3$ **and** $100 \times 5 + 10 \times 0 + 8$

2 Tricks with reversed digits

Operation	Result	Always divisible by
Two-digit number $+$ its reverse	$11(a + b)$	11; quotient is the sum of the digits
Two-digit number $-$ its reverse	$9(a - b)$	9; quotient is the difference of the digits
Three-digit number $-$ its reverse	$99(a - c)$	99; quotient is the difference of the end digits
$abc + bca + cab$	$111(a + b + c)$	111, and so by 37 and 3

Example 2.1 Without adding, find the quotient when $58 + 85$ is divided by 11.

SOLUTION

The sum is $11(5 + 8)$.

13

Example 2.2 Without subtracting, find the quotient when $742 - 247$ is divided by 99.

SOLUTION

The difference is $99(7 - 2)$.

5

Example 2.3 The sum of a two-digit number and the number formed by reversing its digits is 121. The digits differ by 3. Find the number.

SOLUTION

$11(a+b)=121$ gives $a+b=11$; with $a-b=3$ we get $a=7$, $b=4$.

74 or 47

3 Letters for digits

- In these puzzles each letter stands for **one digit**, the same letter always stands for the same digit, and the leading digit of a number is not zero.
- Work column by column from the units place, keeping track of the **carry**.

Example 3.1 Find A and B: 3A + 25 = B2 (two-digit numbers).

SOLUTION

Units: $A+5$ ends in 2, so $A=7$ with carry 1.

Tens: $3+2+1=6$, so $B=6$.

$37+25=62$

Example 3.2 Find A and B: A1 × 3 = 9B (A1 and 9B are two-digit numbers).

SOLUTION

Units: $1\times 3=3$, so $B=3$. Tens: $3A=9$, so $A=3$.

$31\times 3=93$

Example 3.3 Find A, B and C: AB × 5 = CAB, where AB is a two-digit number and CAB a three-digit number.

SOLUTION

$5\times B$ ends in B, so $B=0$ or $B=5$. Try $B=5$: $25\times 5=125$ and $75\times 5=375$ both work; with $B=0$, $50\times 5=250$ works too.

$25\times 5=125$, $75\times 5=375$ **or** $50\times 5=250$

4 Tests of divisibility

Divisor	Test
2	The units digit is 0, 2, 4, 6 or 8
3	The sum of the digits is divisible by 3
4	The number formed by the last two digits is divisible by 4
5	The units digit is 0 or 5
6	Divisible by both 2 and 3
8	The number formed by the last three digits is divisible by 8
9	The sum of the digits is divisible by 9
10	The units digit is 0
11	(Sum of the digits in odd places) − (sum of the digits in even places) is 0 or a multiple of 11

- **Why the tests for 2, 5 and 10 work**: $10a + b$ differs from b by a multiple of 10, so only the units digit matters.
- **Why the tests for 3 and 9 work**: $100a + 10b + c = 99a + 9b + (a + b + c)$. The first part is divisible by 9, so the number and its digit sum leave the same remainder.
- **Why the test for 11 works**: $100a + 10b + c = 99a + 11b + (a - b + c)$.
- For other divisors use **co-prime factors**: 12 uses 3 and 4; 15 uses 3 and 5; 18 uses 2 and 9; 24 uses 3 and 8; 45 uses 5 and 9.

Example 4.1 Test 73,548 for divisibility by 2, 3, 4, 6, 8 and 9.

SOLUTION

Units digit 8: divisible by 2. Digit sum 27: divisible by 3 and 9. So also by 6.

Last two digits 48: divisible by 4. Last three digits 548: $548 = 8 \times 68 + 4$, not divisible by 8.

Divisible by 2, 3, 4, 6 and 9; not by 8

Example 4.2 Is 9,18,082 divisible by 11?

SOLUTION

Odd places from the right: $2 + 0 + 1 = 3$. Even places: $8 + 8 + 9 = 25$. Difference 22.

Yes

5 Finding missing digits

- Use the test for the given divisor to form a condition on the missing digit.

Example 5.1 The number $21y5$ is a multiple of 9. Find the digit y.

SOLUTION

Digit sum $= 8 + y$ must be a multiple of 9.

$y = 1$

Example 5.2 The number $31z5$ is a multiple of 3. Find the possible values of z.

SOLUTION

$9 + z$ must be a multiple of 3.

$z = 0, 3, 6$ **or** 9

Example 5.3 Find the digit x if $4x32$ is divisible by 11.

SOLUTION

Odd places: $2 + x$. Even places: $3 + 4 = 7$. So $x - 5$ is 0 or a multiple of 11.

$x = 5$

Example 5.4 Find the digits a and b if $7a8b$ is divisible by 45 and $b \neq 0$.

SOLUTION

Divisible by 5 with $b \neq 0$: $b = 5$. Divisible by 9: $20 + a$ is a multiple of 9.

$a = 7, b = 5$

6 Common mistakes

- Treating ab as $a \times b$ when it stands for a two-digit number.
- Testing divisibility by 4 with only the units digit.
- Testing divisibility by 12 with 2 and 6. Use 3 and 4, which have no common factor.
- For 11, adding all the digits instead of taking alternate sums.
- Giving only one value when several digits satisfy the condition.
- Forgetting the carry in letter puzzles.